

A video-based approach to predicting roulette

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Thorp & Shannon

- ▶ Edward Thorp & Claude Shannon
- ▶ First wearable computer (1961)
- ▶ Measured ball/wheel velocities
- ▶ 44% edge over house



Physics-based approach using
human input

Thorp, E. O. (1998). The invention of the first wearable computer. *IEEE Symposium on Wearable Computers*.

More recent research:

Small, M. & Tse, C. K. (2012)

"Predicting the outcome of roulette" - Chaos Journal

- ▶ Deterministic chaos model of roulette dynamics
- ▶ Calculate an expected 18% edge on a perfect wheel

Kanhirun, P. et al. (2008)

"Neural network model for roulette" - IEEE ISCIT

- ▶ Image processing to estimate parameters
- ▶ Achieves 40% returns

Q: Can we beat roulette using an end-to-end deep learning approach?

Project Overview

1. **Dataset Collection & Preprocessing**
2. **Model Architecture Design**
 - ▶ ResNet3D-18 with transfer learning
3. **Training & Optimization**
 - ▶ Class balancing, fine-tuning
4. **Evaluation & Results**
 - ▶ Accuracy metrics and betting strategy analysis

Dataset: RouletteVision

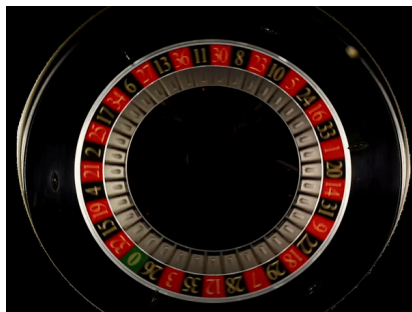
RouletteVision Dataset

Each video split into two parts:

Input: Ball spinning around the wheel (2-5+ seconds)

Output: Landing pocket label

1,703 samples



Dataset:

<https://huggingface.co/datasets/mp-coder/RouletteVision-Dataset>

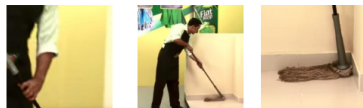
Model Architecture

R3D-18 (3D ResNet-18)

- ▶ Video classification model
- ▶ 3D convolutions for spatiotemporal learning
- ▶ Pretrained on Kinetics-400

Why is this suitable?

- ▶ Trained to recognize motion patterns
- ▶ Understands temporal dynamics
- ▶ Provides feature extraction relevant to motion



Ground Truth: Mopping Floor, Result: Mopping Floor



Ground Truth: Decorating the Christmas Tree, Result: Decorating the Christmas Tree

Kinetics-400: 400 human action classes

Training Strategy

1. Preprocessing

- ▶ Clip to only first 16 frames of input
- ▶ Resize and normalize

2. Transfer Learning Setup

- ▶ Freeze pre-trained backbone
- ▶ Replace final classifier with custom head
- ▶ Dropout $\rightarrow$ Linear(512 $\rightarrow$ 256) $\rightarrow$ ReLU $\rightarrow$ Linear(256 $\rightarrow$ 37)

3. Fine-Tuning

- ▶ Unfreeze final ResNet layer
- ▶ Lower learning rate for pretrained weights

Challenges

▶ **Class Imbalance**

- ▶ Slight imbalance in dataset
- ▶ Weighted sampling to balance training

▶ **Limited Data Per Class**

- ▶ 1,703 videos across 37 classes 46 per number
- ▶ Risk of overfitting to specific wheel/camera conditions

▶ **Statistical Significance**

- ▶ Any edge over random chance/break even will be small
- ▶ Dataset is all of one table, in an uncertain environment

Betting Strategy: Top-5

The Strategy

- ▶ Model outputs probability distribution over all 37 numbers
- ▶ Bet 1 unit each on the 5 most likely numbers

Roulette Payout

- ▶ Single number bet pays 35:1
- ▶ Win = 1 original bet + 35 profit = 36 total return

Break-Even Analysis

- ▶ Need to hit one of the top-5 numbers to win
- ▶ Expected return = Top-5 Accuracy $\times$ 36
- ▶ Break-even: $\frac{5 \text{ units cost}}{36 \text{ total return}} = 13.89\% \text{ accuracy}$
- ▶ Random guessing: $\frac{5}{37} = 13.51\%$

Need $\geq 13.89\%$ top-5 accuracy to break even

Results

- ▶ **Top-1 Betting:** +12.0% player edge
 - ▶ Accuracy: 3.12% (beats 2.78% break-even)
- ▶ **Top-5 Equal Betting:** +6.8% player edge
 - ▶ Accuracy: 14.84% (beats 13.89% break-even)
- ▶ **Top-5 Proportional Betting:** +8.4% player edge
 - ▶ Splitting bet amounts based on confidence in a pocket

Standard roulette house edge: -2.7%

Questions